

Lab Record

Question 1 — Insertion Sort

Input:

Enter number of elements: 5

Enter elements: 12 5 8 20 3

Output:

Sorted elements using Insertion Sort: 3 5 8 12 20

10 15 20 25 30

Question 2 — Merge Sort

Input:

Enter number of elements: 7

Enter elements: 8 3 5 2 9 7 1

Output:

Sorted elements using Merge Sort: 1 2 3 5 7 8 9

Question 3 — Shell Sort

Input:

Enter number of elements: 6

Enter elements: 45 12 78 23 56 10

Output:

Sorted elements using Shell Sort: 10 12 23 45 56 78

Question 4 — Radix Sort

Input:

Enter number of elements: 8

Enter elements: 73 44 25 99 35 12 65 89

Output:

Sorted elements using Radix Sort: 12 25 35 44 65 73 89 99

Question 5 — Hashing (Linear Probing)**Input:**

Enter table size: 7

Enter 5 elements: 20 36 41 58 25

Output:

Hash Table (Linear Probing):

Index 0 → 41

Index 1 → 58

Index 2 → 36

Index 3 → 25

Index 4 → 20

Index 5 → —

Index 6 → —
